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Original Article

Conceptual Design of a Turboprop Counter-Insurgency Aircraft: "JARZ-01"

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In the face of rising global tensions and insurgent threats, a dedicated aircraft is designed specifically for counter-insurgency missions. Equipped with short takeoff and landing (STOL) capabilities on unpaved runways, versatile weaponry, and a focus on survivability and cost-effectiveness, this specialized aircraft addresses the unique challenges of modern conflict. This paper explores the conceptual design of the JARZ-01 Counter-Insurgency (COIN) military-grade turboprop aircraft, that is meticulously crafted to excel in light attack, surveillance, and ferry operations. Employing analytical calculations, software simulations, and insights from references, various configurations are assessed in detail. In conclusion, the study presents the conceptual design of the JARZ-01 COIN aircraft, demonstrating its compliance with the Design Requirements and Objectives (DRO).

Keywords: counter-insurgency (COIN), conceptual design, military-grade turboprop aircraft

1. Introduction

In the ever-changing global landscape, safeguarding national sovereignty requires a strong air defense. The effectiveness of air defense relies on reliable equipment and technological self-sufficiency. Geopolitical tensions have prompted the need for modernizing air defense, including the development of Counter-Insurgency (COIN) aircraft.

The ongoing geopolitical tensions worldwide have catalyzed the imperative for the modernization of air defense systems. This extends to the enhancement of air defense equipment, spurred by the heightened activities of insurgency in various regions. The surge in insurgency has given rise to the development of Counter-Insurgency (COIN) aircraft. These aircraft are strategically designed to operate in challenging terrains, necessitating capabilities such as Short Takeoff and Landing (STOL) or Vertical Takeoff and Landing (VTOL) on unpaved runways. Moreover, their weaponry must be versatile enough to address a diverse range of targets.

In response to these dynamic challenges, Indonesia and several allied nations have formed a collaborative consortium. This consortium's primary objective is to develop an advanced COIN aircraft, slated to commence operations by 2030. This joint effort harnesses cutting-edge technology, with an estimated production of approximately 100 aircraft. Beyond addressing defense needs, this initiative is strategically positioned to boost national self-sufficiency, ensuring the nation's security and resilience in the face of emerging threats. This conceptual design not only aims to meet defense requirements but stands as a testament to the collaborative consortium's commitment to pioneering technological innovation and securing the nation's future in the face of evolving challenges.

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2. Material and Methods

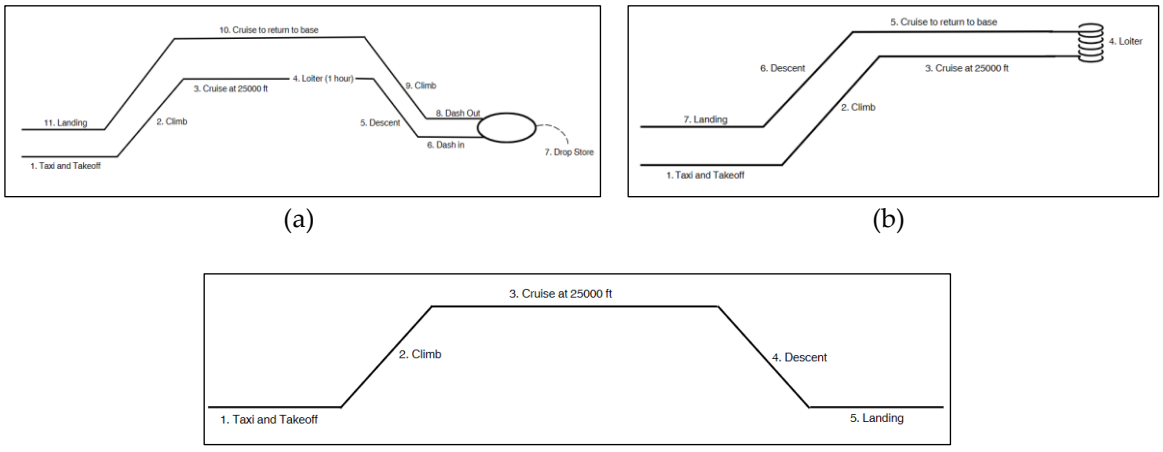
2.1. Design Requirements and Objectives (DRO)

The new COIN aircraft is being designed based on the specified Design Requirements and Objectives (DRO). The DR&O of JARZ-01 is shown in Table 1.

Table 1. JARZ-01 Design Requirement & Objectives

REQUIREMENT			
General		**Flight Performance**	
Aircraft Type	Capable of light attack, ferry, and surveillance operations	Radius of Action	≥ 350 km
Load Factor	+6/-3	Loiter Time	≥ 60 min
Weapons and Payload		Dash in/Dash Out Distance	≥ 50 km
Weight	≥ 1550 kg	Service Ceiling	≥ 30000 ft
Number of Hardpoints	≥ 5	Takeoff/Landing Distance (85% MTOW)	≤ 3600 ft
Flight Decks and Instruments		Rate of Climb (70% MTOW)	≥ 3500 ft/min
Design	Design for 2 pilot crew members	**Airworthiness Certification**	
Avionics	The ability to carry air-to-ground ammunitions.	Certification Basis	MIL-SPECS
OBJECTIVES			
General		**Flight Performance**	
Cost	Competitive Price	Stall Speed (70% MTOW Sea Level)	< 135 km/h
		Maximum Speed	550 km/h
		Ferry Range	2500 km

With the DR&O specified, the JARZ-01 will have three roles: light attack, surveillance, and ferry. The mission profile for all of the roles is shown in Figure 1.



(a) (b)

(c)

Figure 1. JARZ-01 Mission Profile: (a) Light Attack; (b) Surveillance; (c) Ferry

e-ISSN: 2715-6958 p-ISSN: 2721-5342

2.2. *Design Methods*.

The aircraft design process is iterative, beginning with a detailed assessment of the DRO and a thorough comparison with similar aircrafts. This leads to the determination of the preliminary design, focusing on key elements such as shape, size, and dimensions. Configuration decisions involve careful consideration, followed determining the design point that aligns with the DRO.

Following the establishment of the initial design, a meticulous analysis and iteration phase ensue. Comprehensive examinations of aerodynamics, stability, structure, and performance ensure alignment with the DRO, with identified discrepancies prompting necessary design adjustments and reassessments.

Aerodynamic and stability analysis will be performed using XFLR and Digital DATCOM software to obtain the necessary aerodynamic and stability coefficients and data.

The structural analysis is performed using data from Roskam [1]. The structural analysis will encompass the details of the aircraft's spars, ribs, longerons, frames and skin panels.

Aircraft performance is crucial for quantitatively validating its capabilities. It is anticipated that the aircraft performance will surpass the Design Requirements and Objectives (DR&O). This article calculates various performance metrics such as takeoff distance, landing distance, rate of climb, service ceiling, range, and endurance. These metrics assess the JARZ-01's ability to fulfill the mission outlined in the DRO. The methods used will be based on Sadraey [2].

For cost analysis, the projected cost of the aircraft is derived from Roskam [1] and computed using the Excel spreadsheet calculation developed by Pasaribu, H. This analysis aims to ascertain the pricing of the aircraft and make comparisons with competing aircraft.

3. Results

3.1. *Design Point and Engine Selection*

The design point of the aircraft must be chosen to determine the required wing area and the engine power. The design point will be determined using the matching chart. The matching chart shown in Figure 2, will show each requirement's power loading (W/P) and wing loading (W/S).

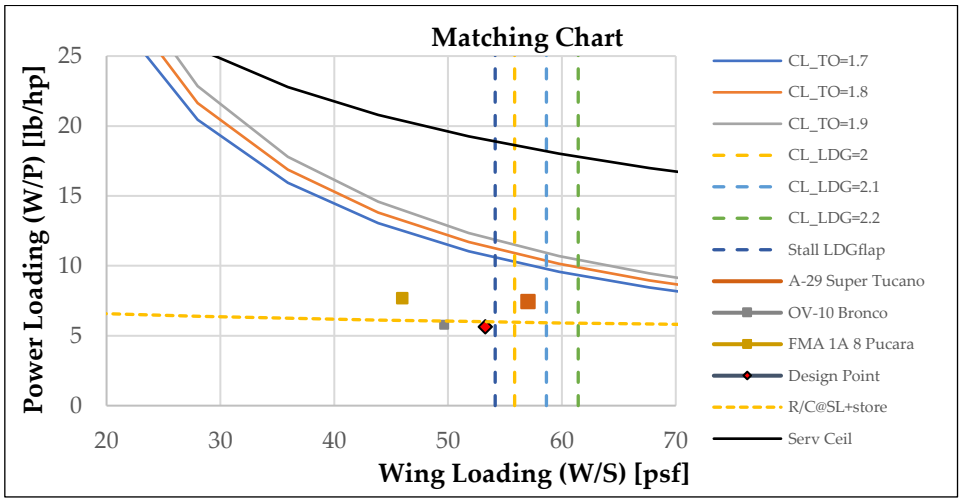


Figure 2. JARZ-01 Matching Chart

Based on Figure 2, the design point of JARZ-01 will be chosen at a wing loading of 53.3 lb/hp and a power loading of 5.6 psf. This will result in a required engine power of 2300 hp and a wing area of 22.6 m^2. From this value, the engine chosen will be the PW127XT-M engine. This a modern turboprop engine developed by Pratt & Whitney Canada (P&WC) in 2021. This particular engine has a power of 2600 hp, this power is greater than the design point due to the fact that propulsive engine

e-ISSN: 2715-6958 p-ISSN: 2721-5342

power accounts for 90% of the total power while the remaining 10% is used for other components in the aircraft (avionics, hydraulics, fuel systems, etc.) as the aircraft does not have an auxiliary power unit. The full specifications of the engine is shown in Table 2.

Table 2. PW127XT-M Engine Specifications

Engine Name	Dry Weight (kg)	SFC (lb/(hp.h))	Length (m)	Diameter (m)	Max Continuous Power (hp)	Max Take-off Power (hp)
PW127XT-M	481	0.445	2.1	0.68	2600	2750

3.2. Aircraft Final Design

The final design of JARZ-01 is shown in Figure 3 below. In addition, the JARZ-01 will be carry several armaments that are attached to the wings and lower fuselage (6 hardpoints in total), adding the weight, the aircraft will carry 1593 kg of payload.

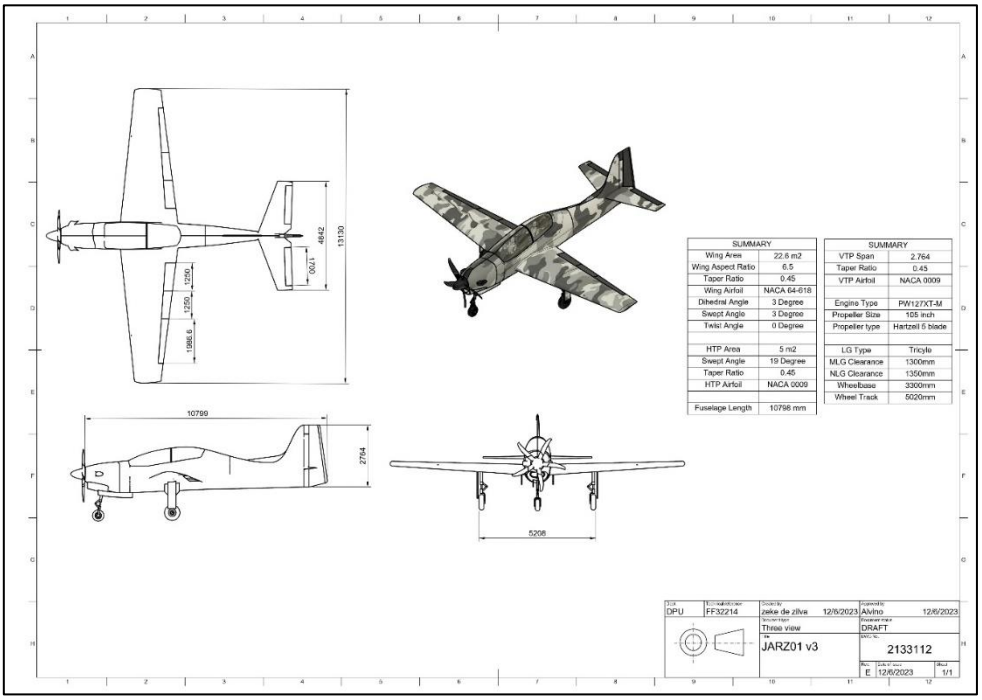


Figure 3. JARZ-01 Three-View Drawing

3.2.1. Fuselage Configuration

The fuselage structure of JARZ-01 will be semi-monocoque and have an oval shape cross-section similar to military fighters as seen in Figure 4. The fuselage will have a total length of 10.8 m, a cockpit length of 3.4 m and a nose length of 2.6 m.

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e-ISSN: 2715-6958 p-ISSN: 2721-5342

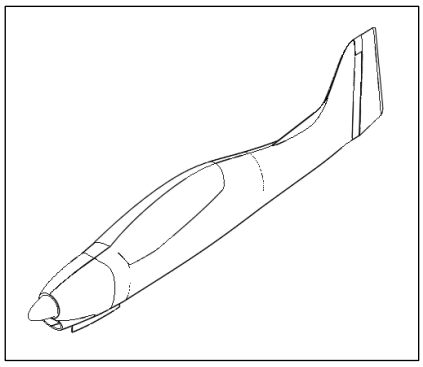


Figure 4. JARZ-01 Fuselage

The cockpit design of the JARZ-01 can be seen in Figure 5. and it is designed that it can fit two pilots in a tandem configuration and both pilots have sufficient visibility to fly the aircraft.

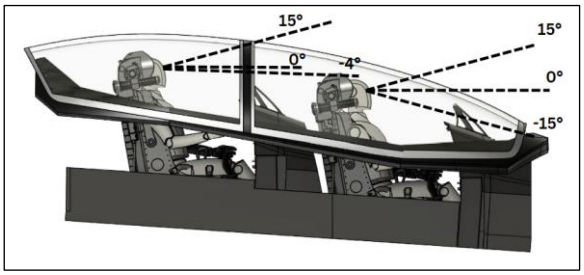


Figure 5. JARZ-01 Cockpit Design

3.2.2. Wing Configuration

The wing design of JARZ-01 is chosen based on the advantages and disadvantages on the reference [3-4] and comparing similar aircrafts. After thorough analysis, the wing specifications of JARZ-01 is shown in Table 3. The JARZ-01 wing will have a low-wing placement and a tapered wing planform.

Table 3. JARZ-01 Wing Data

Wing Dimensions			**Flap Dimension**		
Parameter	Value	Unit			
Wing Area	22.6	[m^2]	1st Flap Span	1.25	[m]
Aspect Ratio	6.5	[-]	2nd Flap Span	1.25	[m]
Wing Span	12.12	[m]	Chord	0.2	[x/c]
Taper Ratio	0.45	[-]	Flap Deflection (Take-off)	25	[◦]
Root Chord	2.57	[m]	Flap Deflection (Landing)	40	[◦]
Tip Chord	1.16	[m]	Flap Type	Plain Flaps	
Sweep Angle at 0.25c	3	[◦]	**Aileron Dimension**		
Dihedral Angle	3	[◦]			
Twist Angle	0	[◦]	Aileron Span	2.0	[m]
Airfoil	NACA 64-618		Chord	0.2	[x/c]

3.2.3. Empennage Configuration

Similar to the JARZ-01 wing, the JARZ-01 empennage; Horizontal Tail Plane (HTP) and Vertical Tail Plane (VTP); will be based on the reference [3-4]. The empennage will have a conventional

 doi: 10.47355/AVIA.VxIx.XX

e-ISSN: 2715-6958 p-ISSN: 2721-5342

configuration with plain elevators and a rudder in the HTP and VTP respectively. The empennage specifications of JARZ-01 is shown in Table 4.

Table 4. JARZ-01 Empennage Dimensions

Empennage Dimensions			
Parameters	HTP	VTP	Unit
Area	5	3.34	[m^2]
Aspect Ratio	4.33	2.28	[-]
Span	4.7	2.76 (Half-span)	[m]
Taper Ratio	0.45	0.45	[-]
Tip Chord	0.7	0.84	[m]
Root Chord	1.5	1.66	[m]
Sweep Angle at 0.25c	19	0	[°]
Twist Angle	0	0	[°]
Dihedral Angle	0	0	[°]
Airfoil	NACA 0009		

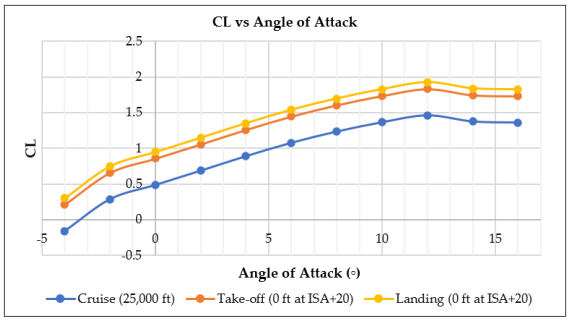
3.3. *Aerodynamic Analysis*

The wing used in JARZ-01 is NACA 64-618 as the wing's cross-sectional area and plain flaps is also used as the high-lift devices with 20° deflection for take-off and 40° for landing. To fulfill the DRO, the aircraft lifting surfaces must satisfy the lift coefficient based on the calculation done for matching chart in Table 5.

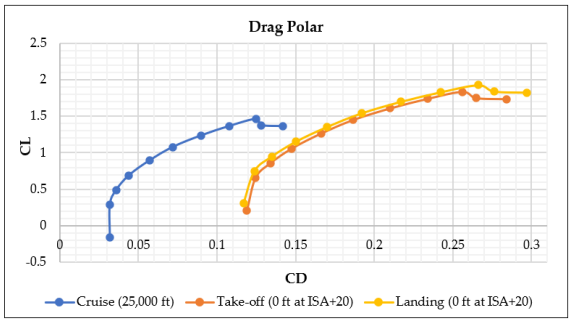
Table 5. Aircraft Maximum Lift Coefficient Requirements

Parameter	Cruise	Take-off	Landing
$C_{L_{max}}$	1.5	1.8	2.1

The analysis on aerodynamics for JARZ-01 is done on three flight phases, namely take-off, landing, and cruise. To analyze the aerodynamic characteristics results of JARZ-01, Digital DATCOM software is used. Extra drag will be added due to landing gear, flaps, armaments, camera, and radar, which will be conducted by using component built-up method based on reference [2]. The calculated aerodynamic characteristics are shown in Figure 6 and the value comparison with the matching chart is shown in Table 6.



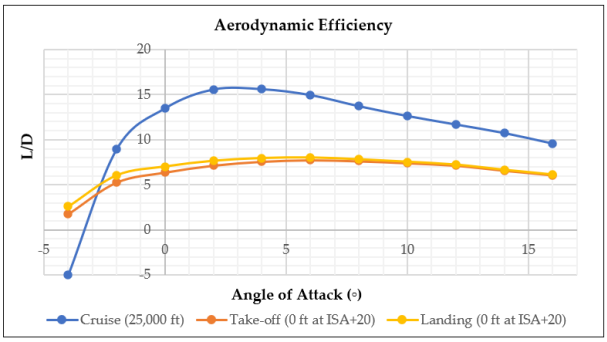
(a)



(b)

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e-ISSN: 2715-6958 p-ISSN: 2721-5342



(c)

Figure 6. Aerodynamic Characteristics of: (a) C_{L_α}; (b) Drag Polar; (c) $(L/D)_\alpha$

Table 6. Comparison of Calculated Aerodynamic Parameters and Matching Chart

Flight Phase	$C_{L_{max}}$		C_{D_0}	
	Calculation	Matching Chart	Calculation	Matching Chart
Cruise	1.56	1.5	0.032	0.022
Take-off	1.83	1.8	0.078	0.06
Landing	2.15	2.1	0.124	0.122

JARZ-01 fulfills the required $C_{L_{max}}$ and C_{D_0} for all three flight phases. JARZ-01 also obtained a maximum value of L/D of 21.44 at an angle of attack of 2°.

3.4. *Aircraft Weight and Balance*

The weight estimation of the JARZ-01 aircraft is determined using Roskam's method [1]. A weight breakdown for the aircraft is presented in Table 7. A comparison was made between obtained values and reference values from Roskam [1] to validate consistency. Results showed that most criteria were met, but this may be due to the trainer role aircraft carrying all payloads and fuel, which is not the case during light attack missions where fuel capacity is not always at its maximum.

Table 7**.** MTOW to Weight Percentage Criteria

Weight Groups	Final Weight			MTOW Percentage Criteria	
	(lb)	(kg)	%TOW	Minimum	Maximum
Wing	1212	550	9.38%	9%	13%
Total Empennage	407	185	3.15%	2%	5%
Fuselage	1069	485	8.27%	6%	15%
Total Landing Gear	513	233	3.97%	4%	6%
Nacelle Group	179	81	1.39%	1%	5%
Airframe Weight	3380	1533	26.16%	20%	40%
Total Propulsion System	1631	740	12.62%	12%	20%
Total Fixed Equipment	1398	634	10.82%	7%	15%
Empty Weight	6408	2907	49.60%	45%	65%

After calculating the weight, the extreme left and right positions are labeled as the most-aft and most-forward CG positions, respectively. These values are 17% and 24%, respectively, of the Mean Aerodynamic Chord. By analyzing the shift at each stage, a more stable aircraft can be designed.

e-ISSN: 2715-6958 p-ISSN: 2721-5342

Figure 7 shows the potato diagram results for each mission role. The payload placement causes the center of gravity to shift left, then right, and then left again. The light attack role experiences a significant shift due to the payload drop, while the surveillance role has a radar located between the fuselage and empennage, which may account for the slight shift.

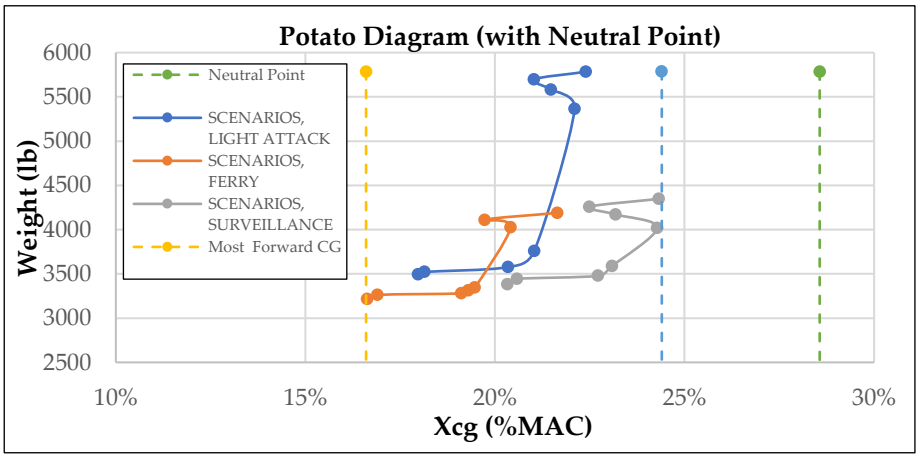


Figure 7. JARZ-01 Potato Diagram

3.5. *Structural Layout*

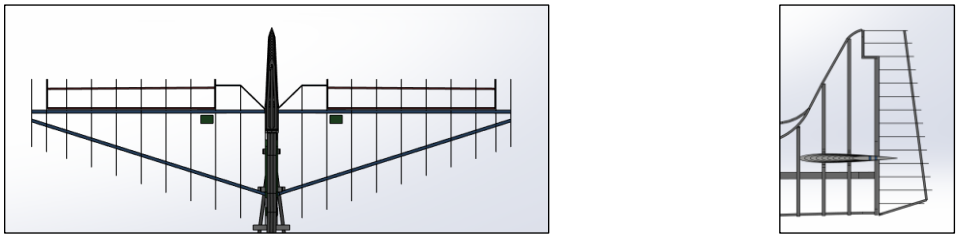
The JARZ-01 aircraft has 21 ribs in each wing, with the largest rib spacing being around 0.3 m. The rib thickness is set at 2.5 mm and 3 mm for hardpoints. There are four hardpoints in the configuration of the JARZ-01 wing, including the wing root as a strategic hardpoint. The upper wing skin is made of Al2024-T3, while Al7075-T6 is used for the lower wing skin and wing ribs. Al7475-T7351 is selected for the stringer. The structure layout is shown in Figure 8.



(a) (b)

Figure 8. Wing Structural Layout: (a) Ribs Placement (b) Landing Gear Placement

The Horizontal Tail Plane (HTP), shown in Figure 9a, will have two main spars located in 25% and 75% chord for the forward and aft spars respectively, with a thickness of 1 mm. The Vertical Tail Plane (VTP), in Figure 9b, is integrated with the fuselage, and the rudder features a singular spar accompanied by 14 ribs, each with a 1 mm thickness. Al 7075-T6 is chosen for the spars and ribs of both the HTP and VTP components while Al 2024-T3 is designated for the skin panel sections.



(a) (b)

Figure 9. Empennage Structural Layout: (a) Horizontal Tail Plane (b) Vertical Tail Plane

e-ISSN: 2715-6958 p-ISSN: 2721-5342

The JARZ-01's fuselage structure has been carefully designed with attention given to the longerons, frames, and bulkheads. The aircraft features 4 longerons, 24 fuselage frames, and 4 bulkheads, each made from carefully selected materials such as Al 2024-T3 for skin panels and Al 7075-T6 for frames, bulkheads, and stiffeners. The spacing between frames and bulkheads is maintained at a minimum of 15 inches, with variations near the nose due to engine and propeller placement. Load-bearing components such as stringers and floor beams are also made from Al7075-T6, ensuring the fuselage structure can handle anticipated loads while maintaining strength and integrity.

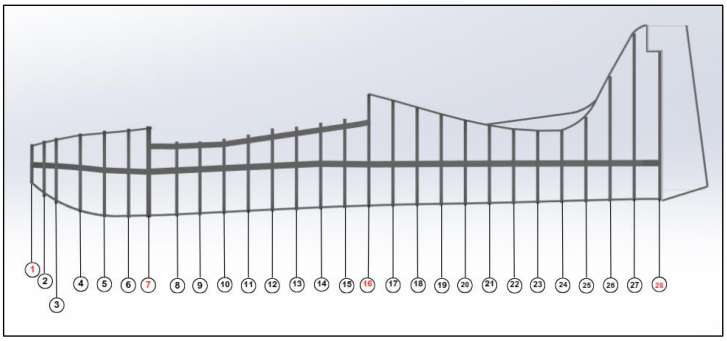


Figure 10. Fuselage Structural Layout

In Figure 10, the JARZ-01 aircraft bulkheads are numbered as 1, 7, 16, and 28. The longerons are placed at 8–12-inch intervals to ensure optimal load distribution, with a thickness of 1 mm each. The frames have a thickness of 50 mm, while the bulkheads have a thickness of 75 mm. Bulkhead no. 1 is located at the junction of the propeller and the fuselage, while bulkheads no. 7 and 16 provide structural support for the cockpit area. Additionally, bulkhead no. 28 serves as the support structure for the rudder in the vertical tail plane.

3.6. Internal Systems

Based on all selected equipment and desired capabilities it is possible to depict all systems into internal system diagram as shown in Figure 11.

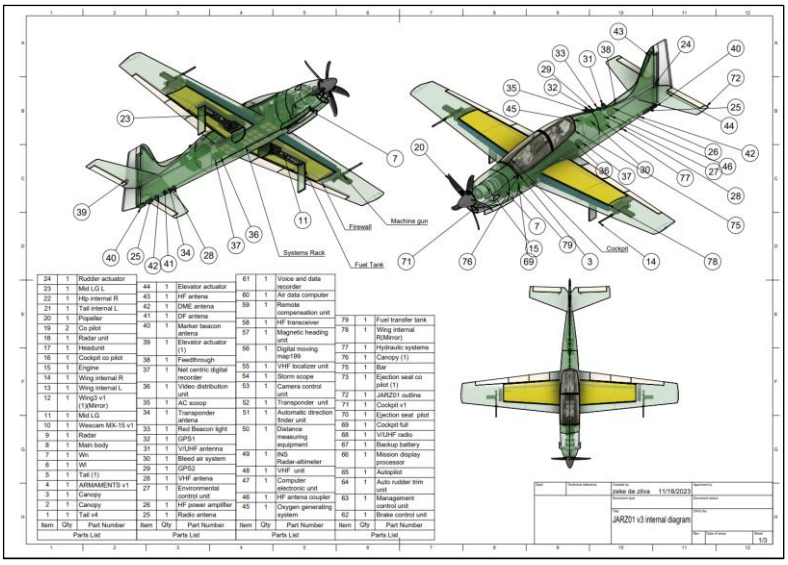
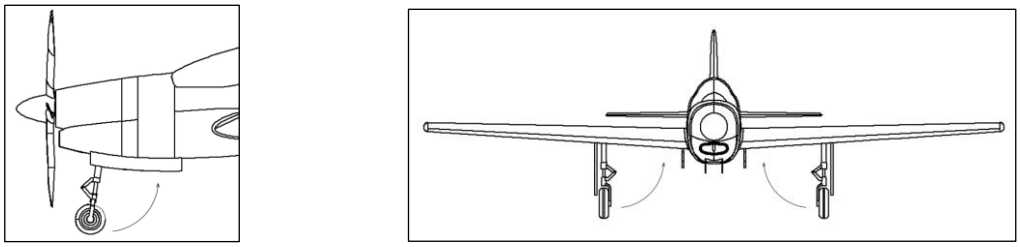


Figure 11. Internal Systems Diagram

e-ISSN: 2715-6958 p-ISSN: 2721-5342

3.7. Landing Gear Design

The JARZ-01 aircraft features a tricycle landing gear configuration with retractable features for superior directional stability on the ground and improved aerodynamic performance during flight. The nose landing gear is strategically located between the cockpit and nose on the bulkhead, maximizing limited space and designed to retract and rotate along the y-axis while fitting seamlessly into the pilot's cockpit. The main landing gear is mounted on the wings due to limited fuselage space, and the retraction mechanism is similar to the nose landing gear but pulled towards the fuselage.



(a) (b)

Figure 12. Landing Gear Retraction: (a) Nose Landing Gear; (b) Main Landing Gear

To be airworthy, there are a couple of criteria that require satisfaction, these are the tip back, overturn and turnover angle. The tip back angle is the maximum nose-up attitude of an aircraft while the tail is on the ground and the landing gear strut is fully extended. It is critical in landing gear design to prevent the aircraft from tipping back on its tail. The landing gear turnover angle plays a critical role in maintaining stability during taxiing and turns, and it is used to prevent tipping during landing and ensure that the aircraft remains upright while on the ground. These values are shown in Table 8.

Table 8. Landing Gear Criteria

Landing Gear Design			**Ground Clearance**		
Criteria	Requirements	JARZ-01	Criteria	Requirements	JARZ-01
Tip back	≥ 15°	17.4°	Longitudinal	≥ 15°	16.1°
Overturn	≤ 55°	38.41°	Lateral	≥ 5°	26.91°
Turnover	≤ 60°	46.91°	NLG	≥ 40°	55.3°

The ground clearance of an aircraft is crucial during takeoff, landing, and ground maneuvers. Longitudinal ground clearance is the angle between the rear of the aircraft and the contact surface of the main landing gear with the ground. The nose landing gear is designed to rotate or tilt for safe takeoff, landing, and ground maneuvers. Lateral ground clearance is the distance between the ground and the aircraft's wingtips. These clearances are important for safe operation during different conditions and terrains. All these criteria have been satisfied as shown in Table 9.

Table 9. Landing Gear Sizing Parameters

Landing Gear Sizing Parameters		
Rated Load (lb)	Main Landing Gear	19904
	Nose Landing Gear	4819
Tire Inflation Pressure (psi)		70-90
Rated Velocity (mph)		133

Table 9 shows the parameters required for JARZ-01. Selecting the most suitable wheel type for an aircraft's landing gear ensures safe and reliable performance during various operational conditions. Factors such as maximum static load, maximum runway speed, and tire pressure are

e-ISSN: 2715-6958 p-ISSN: 2721-5342

considered when determining the optimal size requirements for the tires and wheel. For an aircraft predominantly used for counterinsurgency operations, the ideal tire pressure is determined based on the surface of a small tarmac runway with a good foundation. According to the given parameters, we can determine the specifications for the MLG (Main Landing Gear) and NLG (Nose Landing Gear) tires as shown in Table 10.

Table 10. Landing Gear Specifications

Parameter	Main Landing Gear	Nose Landing Gear
Size	27x8.5	20x7.5
Ply Rating	14	12
Rated Inflation (psi)	90	85
Rated Speed (mph)	165	185
Rated Load (lbs)	11000	5700

3.8. Stability and Control Analysis

The stability analysis for JARZ-01 is calculated using Digital DATCOM software and is shown in Table 11.

Table 11. Static Stability Coefficients of JARZ-01 Aircraft

Parameter	$/deg$	$/rad$	Stability Criteria
C_{L_α}	0.074	4.248	> 0
$C_{m_\alpha}\ (most-forward)$	-0.018	-1.007	< 0
$C_{m_\alpha}\ (most-aft)$	-0.006	-0.353	< 0
C_{Y_β}	-0.011	-0.631	< 0
C_{l_β}	-0.010	-0.593	< 0
C_{n_β}	0.109	6.259	> 0

All the stability parameters meet the stability criterion for JARZ-01 to fly stably. The static margin for the most-forward and most-aft CG 5.75% and 2.01% respectively. The control capacity diagram to determine the optimal size of the HTP is shown in Figure 13.

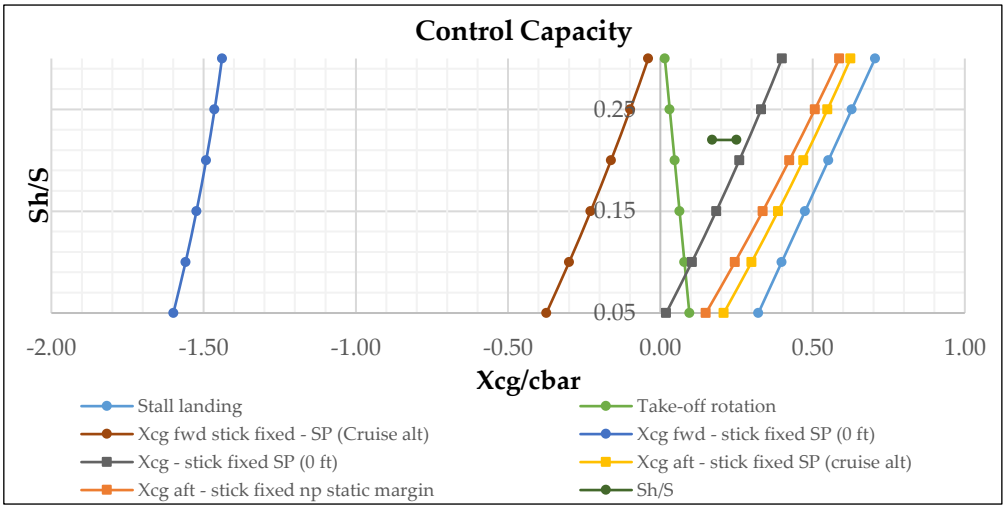


Figure 13. HTP Control Capacity of JARZ-01 Aircraft

e-ISSN: 2715-6958 p-ISSN: 2721-5342

Based on the static margin value, it can be analyzed that JARZ-01 fulfills the static margin requirement of having both values less than 15%. As for the control capacity analysis, it can be seen that the designed HTP area is 0.22 and it satisfies with the control capacity diagram.

3.9. Performance Analysis

The performance of the JARZ-01 is dependent on the aircraft's take-off and landing distances, rate of climb, speed performance, service ceiling, range, and endurance capabilities. These parameters will be based on the power curve of the PW127XT-M engine. This power curve shown in Figure 14, shows the engine's performance at the cruising altitude at 25000 ft.

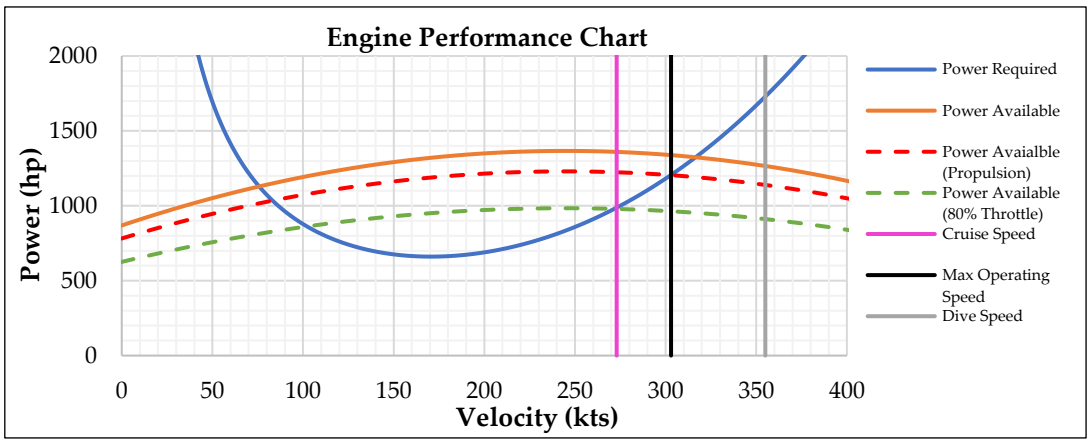


Figure 14. JARZ-01 Engine Performance Chart at Cruise (25000 ft)

As seen in Figure 14, the intersection between the power required and the power available line is the resulted speed. Therefore, the maximum speed of the JARZ-01 aircraft is 304 kts (~563 km/h) and the cruising speed is 273 kts (~505 km/h).

The take-off procedure will be divided into three main phases: ground run ($0 - V_R$), transition ($V_R - V_{LOF}$) and airborne phase ($V_{LOF} - V_2$). The calculation will utilize the kinematic methods of acceleration shown in Equation 1.

$$a = g\left(\bar{T}/_W - \mu_g - \frac{\frac{1}{2}\rho V^2 S\left(C_{D_G} - \mu_g C_{L_G}\right)}{W}\right) \quad (1)$$

The acceleration will be calculated for specified intervals of the aircraft's speed for the first and third phase to obtain the time elapsed, which will be used to calculate the distance. For the transition phase, the time elapsed will be assumed to be 3 seconds [2]. The resulted calculation is shown in Table 13. The take-off distance of JARZ-01 is approximately 523 m or 1715 ft.

Table 12. JARZ-01 Take-off Distance Calculation

Phase	Speed Interval [m/s]	Time [s]	Distance [m]
Ground run	0 – 47.3	13.8	335
Transition	47.3 – 51.8	3.0	149
Airborne	51.8 – 54.1	0.7	39
Total Take-off Distance [m]			523

For the landing, it will be divided four phases: airborne ($V_{app} - V_{flare}$), transition ($V_{flare} - V_{td}$), free-run ($V_{td} - V_B$) and break-stop phase ($V_B - 0$). Using the same method as the take-off calculation,

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the resulted calculation for the landing distance is shown in Table 14. However, it is assumed that approach velocity and flare velocity is the same as the difference between the two speeds is small. The landing distance of JARZ-01 will be 782 m or 2565 ft.

Table 13. JARZ-01 Landing Distance Calculation

Phase	Speed Interval [m/s]	Time [s]	Distance [m]
Airborne	56.6	2.2	124
Transition	56.6 – 47.9	3.0	155
Free-run	47.9 – 35.9	7.3	304
Break-stop	35.9 – 0	11.4	197
Total Landing Distance [m]			782

For the climb performance the power curve in Figure 15 will be used as the maximum rate of climb (RoC) is calculated using the maximum power excess at flight. The results of the maximum rate of climb at various altitude at 75% MTOW are shown in Table 15.

Table 14. Rate of Climb with Variations of Altitude Data (75% MTOW)

Altitude [ft]	Maximum Excess Power [kW]	Maximum RoC [fpm]
0 (Sea Level)	1951	9023
10000	1428	6607
25000 (Cruise)	555	2568
30000	231	1068

The service ceiling is obtained using linear regression of the aircraft's maximum rate of climb at various altitudes. The service ceiling is at the point where aircraft d is unable to climb faster than 100 feet per minute, therefore, the service ceiling of JARZ-01 is 31932 ft. The absolute ceiling is when the aircraft is unable to climb anymore and as for JARZ-01, the absolute ceiling is 32415 ft.

The range and endurance of JARZ-01 will be calculated using the Breguet equation as shown below in equation 2 and 3 respectively.

$$R = \frac{\eta_p}{SFC}\frac{C_L}{C_D} ln\left(\frac{W_1}{W_2}\right) \tag{2}$$

$$E = \frac{2\eta_p}{SFC}\sqrt{S\frac{\rho}{2}\frac{C_L^3}{C_D^2}\left[\frac{1}{\sqrt{W_2}} - \frac{1}{\sqrt{W_1}}\right]} \tag{3}$$

Using the cruising specific fuel consumption (SFC) of the PW127XT-M engine being 0.453 lb/(hp.h) the result of the range and endurance of JARZ-01 for every role is shown in Table 16. To note, the endurance calculation for the light attack mission has taken account of the required 1 hour loiter time. In addition, the radius of action of JARZ-01 during the light attack mission is 413 km.

Table 15. JARZ-01 Endurance and Range for Different Roles

Role	Range [km]	Range [nmi]	Endurance [hrs]
Light Attack	1260	680	3.0
Ferry	4619	2494	13.5
Surveillance	3820	2062	10.9

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Lastly, the Balance Field Length (BFL) of JARZ-01 will only be the take-off distance of the aircraft as the aircraft only has 1 engine and it is not possible to operate under one engine inoperative (OEI). Therefore, the BFL of JARZ-01 is 1715 ft.

3.10. Cost Analysis

To calculate and estimate the cost of an aircraft, there are numerous factors to consider. Due to the complexity of such cost estimation, different methods are used to calculate cost estimation from several references [1-3, 5].Therefor it is possible to graph the breakeven price for minimal 100 units using estimation calculated from the different methods as seen in Figure 15 below

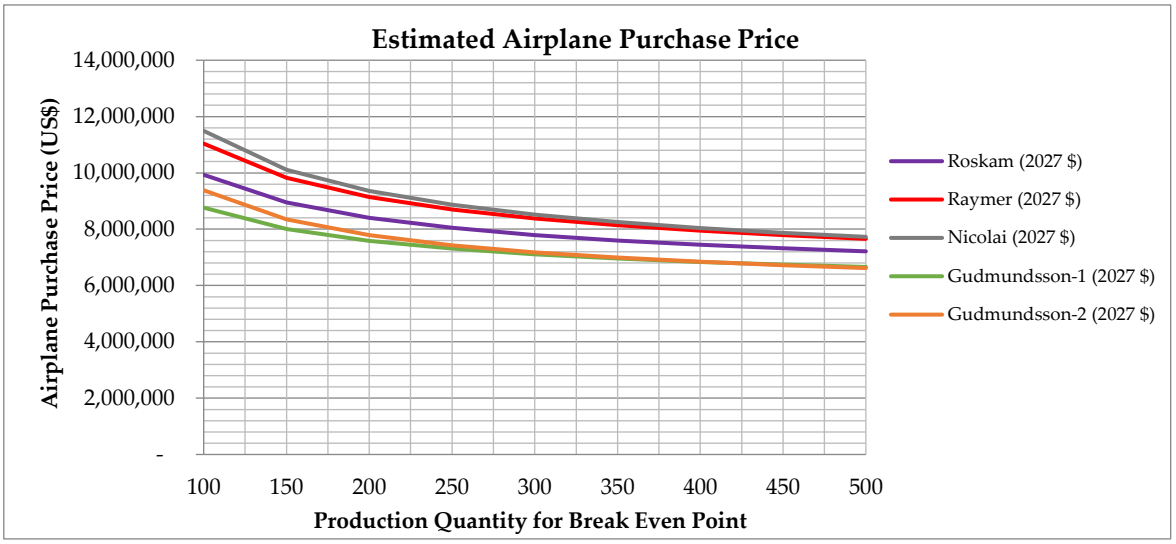


Figure 15. Estimated Airplane Purchase Price Diagram

There for from the graph in Figure 15 it is possible to get an average cost for break-even point of 100 aircrafts, as seen in Table 17.

Table 16. Cost Estimation Price Based On Production Quantity

Production Quantity	Roskam (2030 $)	Gudmundsson-1 (2030 $)	Gudmundsson-2 (2030 $)	Raymer (2030 $)	Nicolai (2030 $)	Average cost
100	9,931,237	8,764,121	9,382,169	11,037,521	11,490,389	**10,121,087**
150	8,952,561	8,008,766	8,347,745	9,823,388	10,112,416	9,048,975
200	8,406,438	7,586,312	7,788,717	9,146,334	9,355,473	8,456,655
250	8,049,059	7,309,389	7,430,883	8,702,718	8,864,849	8,071,380

Based on Table 17 above it is possible to compare the selling price in 2030 with competitor price. As seen in Table 18, it can be concluded that the JARZ-1 is the cheapest and indeed a competitive airplane from cost standpoint for a miliary COIN aircraft.

Table 17. Comparison Price Between Competitors

Reference Aircraft	Price/unit ($)	First flight	Adjusted to 2030
JARZ-1	10.11M (2030)	2030	10.21 M
JARZ-1 Surveillance	13.621 (2030)	2030	13.621 M
A-29 Super Tucano	10.0M (2021)	2003	13.96 M
Fma IA 58 Pucara	4.59 M (1988)	1975	14.68 M
A-37 Dragonfly	1.2M (1965)	1964	14.41 M
T-6 Texan II	8.9M (2007)	2001	16.24 M

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4. Design Review

Based on previous results and analysis, the DRO fulfillment tables of the JARZ-01 is shown in Table 19, this table proves that JARZ-01 has fulfilled the DRO.

Table 18. Design Review Table

Parameter	DRO	Design Value		Fulfillment
Load factor	+6/-2	+6/-2	-	Fulfilled
No. of crews	2	2	-	Fulfilled
Payload weight	≥ 1550	1593	[kg]	Fulfilled
Hardpoints	≥ 5	6	-	Fulfilled
Radius of action	≥ 350	413	[km]	Fulfilled
Loiter time	≥ 60	60	[min]	Fulfilled
Dash in/out distance	≥ 50	50	[km]	Fulfilled
Service ceiling	≥ 30000	31932	[ft]	Fulfilled
Takeoff distance	≤ 3600	1715	[ft]	Fulfilled
Landing distance	≤ 3600	2565	[ft]	Fulfilled
Rate of climb	≥ 3500	9023	[ft/min]	Fulfilled
Maximum speed	≥ 550	563	[km/h]	Fulfilled
Stall speed	≤ 135	135	[km/h]	Fulfilled
Ferry range	≥ 2500	4619	[km]	Fulfilled

5. Conclusions

In conclusion, the conceptual design of the COIN aircraft JARZ-01 has fulfilled all the specified DRO with detailed specification and the design review shown in Table 19 previously. Since this paper only covers the conceptual design, the aircraft can be further optimized by utilizing Computer-Aided Engineering (CAE) Software.

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